

# Temporal Multi-Omics (TMO)

Complete Mathematical Description

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# 1 Notation

- $N$  : number of cells
- $G$  : number of genes
- $P$  : number of ATAC peaks
- $K = 50$  : number of latent components per modality
- $d_{\text{model}} = 64$  : token embedding dimension
- $H = 4$  : number of attention heads
- $L = 2$  : number of transformer layers
- $\tau \in [0, 1]$  : pseudotime
- $Z_{\text{RNA}} \in \mathbb{R}^{N \times K}$  : RNA latent components
- $Z_{\text{ATAC}} \in \mathbb{R}^{N \times K}$  : ATAC latent components
- $a_i(\tau), r_i(\tau)$  : standardised ATAC and RNA component scores
- $\Delta\tau$  : signed regulatory lag (pseudotime units)
- $\Delta\tau_{\text{max}} = 0.5$  : maximum absolute lag
- $\sigma_{\text{min}}^2 = 0.001$  : minimum window width
- $\lambda_{\text{recon}} = 1.0, \lambda_{\text{lag}} = 0.1$  : loss weights
- $\sigma_0^2 = 0.01$  : prior variance (absorbed in loss scaling)

## 2 Data Preprocessing

### 2.1 Gene-level ATAC matrix

Let  $X_{\text{peak}} \in \mathbb{R}_{\geq 0}^{N \times P}$  be the peak-by-cell count matrix. Given a binary peak-to-gene annotation matrix  $M \in \{0, 1\}^{P \times G}$ , where  $M_{p,g} = 1$  if peak  $p$  is assigned to gene  $g$ , the gene-level ATAC matrix is

$$A_{\text{gene}} = X_{\text{peak}} M \in \mathbb{R}_{\geq 0}^{N \times G}. \quad (1)$$

$A_{\text{gene}}$  is stored as a sparse CSR matrix in `.obsbm['ATAC_gene']`. Raw RNA counts remain in `.X`.

### 2.2 Pseudotime

RNA counts are normalised to  $10^4$  total reads per cell,  $\log(1 + x)$ -transformed, and the top 2000 highly variable genes are selected. After PCA (30 components) and  $k$ -NN graph ( $k = 15$ ), diffusion maps are computed (`scanpy.tl.diffmap`). The first diffusion component is min-max scaled to  $[0, 1]$  and taken as pseudotime  $\tau$ .

If an early marker gene (e.g. *Pax6*, *CD34*) is provided, its expression vector  $x_m$  is correlated with the initial pseudotime  $\tau_0$ :

$$r = \text{Pearson}(x_m, \tau_0).$$

If  $r < 0$ , the pseudotime is reversed:  $\tau = 1 - \tau_0$ , and a flag `pseudotime_reversed` is stored. In all downstream steps, if the flag is true, the sign of every  $\Delta\tau$  is flipped so that

**positive  $\Delta\tau$  always means ATAC leads RNA (priming).**

### 3 Latent Representations

#### 3.1 RNA components

$$Z_{\text{RNA}} = \text{PCA}_{50}(\log(1 + X_{\text{RNA}})) \in \mathbb{R}^{N \times K}.$$

#### 3.2 ATAC components

$$X_{\text{tfidf}} = A_{\text{gene}} \text{diag}(\log \frac{N}{\mathbf{d}}), \quad d_j = \#\{c : A_{\text{gene}}(c, j) > 0\}. \quad (2)$$

Each row (cell) of  $X_{\text{tfidf}}$  is  $\ell_2$ -normalised. Then truncated SVD yields

$$Z_{\text{ATAC}} = \text{TruncatedSVD}_{50}(X_{\text{tfidf}}) \in \mathbb{R}^{N \times K}.$$

A cell is represented by a sequence of  $2K = 100$  tokens: the first  $K$  tokens are the ATAC components, the remaining  $K$  tokens are the RNA components.

## 4 Ground-Truth Regulatory Lag from CCF

For each matched component pair  $(i, i)$ , we estimate a signed, time-resolved lag surface along pseudotime.

#### 4.1 Interpolation and cross-correlation

The pseudotime axis is non-uniform. Within a sliding window of half-width  $w = 0.1$  centred at  $\tau_0 \in \{0.1, 0.2, \dots, 0.9\}$ , the ATAC and RNA series are linearly interpolated onto a dense regular grid of 200 points. The cross-correlation is computed with `scipy.signal.correlate` in full mode:

$$\text{CCF}_i(\tau_0, \ell) = \frac{1}{|I|} \sum_{t \in I} a_i(\tau_t) r_i(\tau_t + \ell),$$

where  $I = \{t : |\tau_t - \tau_0| \leq w\}$  and  $\ell$  is the lag in pseudotime units. The peak lag is

$$\Delta\tau_i(\tau_0) = \arg \max_{\ell \in [-0.3, 0.3]} \text{CCF}_i(\tau_0, \ell).$$

#### 4.2 Gaussian process smoothing

The raw surface  $\Delta\tau_i(\tau)$  is smoothed with a Gaussian process (Matern-3/2 kernel, length-scale bounds  $[10^{-2}, 0.5]$ , noise-level bounds  $[10^{-4}, 10^0]$ ). The training target for component  $i$  is the smoothed value at the last window:

$$\Delta\tau_i^{\text{target}} = \Delta\tau_i^{\text{smooth}}(\tau = 0.9). \quad (3)$$

If the pseudotime was reversed, the whole surface is multiplied by  $-1$ , so that the target has the correct biological sign.

## 5 Tokenisation

Each component’s scalar value  $v$  is projected to  $d_{\text{model}}$  via a learned linear layer  $\text{proj} : \mathbb{R}^1 \rightarrow \mathbb{R}^{d_{\text{model}}}$ . The token embedding for a component with latent value  $v$ , modality  $m$  ( $m = 0$  for RNA,  $m = 1$  for ATAC), and pseudotime  $\tau$  is

$$\mathbf{x} = \text{proj}(v) + \mathbf{e}_{\text{mod}}(m) + \mathbf{e}_{\tau}(\tau), \quad (4)$$

where

- $\mathbf{e}_{\text{mod}} \in \mathbb{R}^{d_{\text{model}}}$  is a learned modality embedding,
- $\mathbf{e}_{\tau}(\tau)$  is the output of a 2-layer MLP  $\mathbb{R}^1 \rightarrow \mathbb{R}^{128} \rightarrow \mathbb{R}^{64} \rightarrow \mathbb{R}^{d_{\text{model}}}$  (ReLU activations).

The same pseudotime embedding is shared by all tokens of a cell. The sequence of  $2K$  tokens is formed as

$$\mathbf{X} = [\mathbf{x}_{\text{ATAC},1}, \dots, \mathbf{x}_{\text{ATAC},K}, \mathbf{x}_{\text{RNA},1}, \dots, \mathbf{x}_{\text{RNA},K}] \in \mathbb{R}^{2K \times d_{\text{model}}}.$$

## 6 Transformer Encoder

The encoder consists of  $L = 2$  identical TMO blocks. Each block contains:

### 6.1 Within-modality self-attention

Standard multi-head self-attention (no bias) is applied separately to the ATAC tokens and to the RNA tokens.

### 6.2 Asymmetric cross-modal attention

Cross-attention from one modality to the other is biased by the predicted lag. The mechanism is described in section 8.

### 6.3 Feed-forward network

A 2-layer MLP with expansion factor 4:

$$\text{FFN}(\mathbf{x}) = \mathbf{W}_2 \text{GELU}(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2,$$

followed by dropout and residual connections.

Each sub-layer is preceded by LayerNorm and followed by a residual connection.

## 7 Lag and Width Prediction

The cell embedding  $\mathbf{z}_c$  is obtained by mean-pooling the ATAC token outputs of the first (unbiased) pass:

$$\mathbf{z}_c = \frac{1}{K} \sum_{i=1}^K \mathbf{h}_{\text{ATAC},i}^{(1)} \in \mathbb{R}^{d_{\text{model}}}.$$

For each component index  $i$  (separate for ATAC and RNA), a learnable component embedding  $\mathbf{e}_{\text{comp}}(i) \in \mathbb{R}^{d_{\text{model}}}$  is concatenated with  $\mathbf{z}_c$  and fed to two shared MLPs.

### 7.1 LagMLP

$$\mathbf{h}_1 = \text{LayerNorm}([\mathbf{z}_c; \mathbf{e}_{\text{comp}}(i)]), \quad (5)$$

$$\mathbf{h}_2 = \text{GELU}(\text{Linear}_{2d_{\text{model}} \rightarrow d_{\text{model}}}(\mathbf{h}_1)), \quad (6)$$

$$\mathbf{h}_3 = \text{Dropout}_{0.1}(\mathbf{h}_2), \quad (7)$$

$$\mathbf{h}_4 = \text{GELU}(\text{Linear}_{d_{\text{model}} \rightarrow d_{\text{model}}/2}(\mathbf{h}_3)), \quad (8)$$

$$\mathbf{h}_5 = \text{Dropout}_{0.1}(\mathbf{h}_4), \quad (9)$$

$$\hat{\Delta\tau}_{\text{raw}} = \text{Linear}_{d_{\text{model}}/2 \rightarrow 1}(\mathbf{h}_5), \quad (10)$$

$$\hat{\Delta\tau} = \Delta\tau_{\text{max}} \cdot \tanh(\hat{\Delta\tau}_{\text{raw}}). \quad (11)$$

## 7.2 WidthMLP

Identical architecture to LagMLP, except the final activation:

$$\hat{\sigma}^2 = \sigma_{\min}^2 + \text{Softplus}(\text{MLP}_{\text{Width}}(\mathbf{z}_c, \mathbf{e}_{\text{comp}}(i))).$$

The predictors are applied to all  $K$  ATAC and  $K$  RNA indices, yielding

$$\hat{\Delta\tau}_{\text{ATAC}}, \hat{\sigma}_{\text{ATAC}}^2 \in \mathbb{R}^{N \times K}, \quad \hat{\Delta\tau}_{\text{RNA}}, \hat{\sigma}_{\text{RNA}}^2 \in \mathbb{R}^{N \times K}.$$

## 8 Asymmetric Attention Bias

In the second forward pass, cross-attention logits are additively biased.

### 8.1 ATAC $\rightarrow$ RNA

For an ATAC token at pseudotime  $\tau_i$  attending to an RNA token at pseudotime  $\tau_j$ ,

$$b_{ij} = - \frac{(\tau_j - \tau_i - \hat{\Delta\tau}_{\text{RNA},j})^2}{2\hat{\sigma}_{\text{RNA},j}^2}. \quad (12)$$

### 8.2 RNA $\rightarrow$ ATAC

For the reverse direction,

$$b_{ji}^{\text{rev}} = - \frac{(\tau_i - \tau_j + \hat{\Delta\tau}_{\text{ATAC},i})^2}{2\hat{\sigma}_{\text{ATAC},i}^2}. \quad (13)$$

The biases are added to the scaled dot-product attention scores before softmax, making the attention mechanism explicitly asymmetric. Within-modality attention (ATAC-ATAC, RNA-RNA) uses unbiased attention.

### 8.3 Windowed sparse attention (optional)

For large datasets, attention can be restricted to token pairs satisfying

$$|\tau_j - \tau_i - \hat{\Delta\tau}| \leq 3\hat{\sigma}.$$

This reduces the complexity from  $\mathcal{O}(N^2L^2)$  to  $\mathcal{O}(NLW)$  where  $W$  is the constant window size.

## 9 Decoder

After the second pass, the  $K$  RNA token outputs are linearly projected to reconstruct the RNA latent components:

$$\hat{\mathbf{R}} = \text{Linear}_{d_{\text{model}} \rightarrow 1}(\mathbf{H}_{\text{RNA}}^{(2)}) \in \mathbb{R}^{N \times K}.$$

## 10 Training Loss

The model is trained end-to-end on the full dataset with

$$\mathcal{L}_{\text{total}} = \lambda_{\text{recon}} \mathcal{L}_{\text{recon}} + \lambda_{\text{lag}} \mathcal{L}_{\text{lag}}, \quad (14)$$

where  $\lambda_{\text{recon}} = 1.0$ ,  $\lambda_{\text{lag}} = 0.1$ .

### 10.1 Reconstruction loss

$$\mathcal{L}_{\text{recon}} = \frac{1}{NK} \sum_{c=1}^N \sum_{k=1}^K (\hat{\mathbf{R}}_{c,k} - Z_{\text{RNA},c,k})^2. \quad (15)$$

### 10.2 Lag consistency loss

For each ATAC component  $i$ , the predicted lag is averaged over cells:  $\bar{\Delta\tau}_i = \frac{1}{N} \sum_c \hat{\Delta\tau}_{\text{ATAC}}(c, i)$ . The lag loss is

$$\mathcal{L}_{\text{lag}} = \frac{1}{K} \sum_{i=1}^K (\bar{\Delta\tau}_i - \Delta\tau_i^{\text{target}})^2, \quad (16)$$

where  $\Delta\tau_i^{\text{target}}$  is defined in (3).

### 10.3 Symmetric baseline

A symmetric baseline is trained with the identical architecture but with `use_bias = False` (no attention bias) and  $\lambda_{\text{lag}} = 0$  (no lag loss). The baseline therefore only minimises  $\mathcal{L}_{\text{recon}}$ .

### 10.4 Optimisation

Adam optimiser ( $\eta = 10^{-3}$ , weight decay = 0.01) with a stepwise learning-rate scheduler ( $\gamma = 0.5$  at epochs 20 and 40). Batch size 32, 50 epochs. Every 5 epochs the Lag Concordance Score (LCS) is computed on the full dataset (in-set) and the best checkpoint is saved.

## 11 Lag Concordance Score (LCS)

The LCS is the Spearman rank correlation between the per-component mean predicted lags and the CCF-derived targets:

$$\text{LCS} = \rho_{\text{Spearman}}(\{\bar{\Delta\tau}_i\}_{i=1}^K, \{\Delta\tau_i^{\text{target}}\}_{i=1}^K).$$

NaN targets are excluded.  $\text{LCS} > 0.9$  indicates that the model correctly orders the components by their regulatory lag. The LCS is reported as an in-set diagnostic; independent biological validation is provided by ChIP-seq and Perturb-seq experiments.

## 12 ChIP-seq Validation

For a transcription factor with top-500 bound genes, component-level CCF lags  $\Delta\tau^{\text{CCF}} \in \mathbb{R}^K$  are projected to gene-level lags via the PCA loadings  $\mathbf{L} \in \mathbb{R}^{G \times K}$ :

$$\text{gene\_lag} = \tilde{\mathbf{L}}^\top \Delta\tau^{\text{CCF}},$$

where  $\tilde{\mathbf{L}}$  is obtained by row-normalising  $|\mathbf{L}|$  so that each row sums to 1.

The full set of non-target genes (optionally restricted to those within  $\pm 0.5$  log-expression units of the target-gene median) is used as background. A two-sided Mann-Whitney  $U$  test compares the  $\Delta\tau$  values of target and background genes. This test is deterministic (no random subsampling) and independent of the model predictions.

## 13 Perturb-seq Validation

For a knockout (e.g. SMARCB1), the asymmetric model is trained on non-targeting control cells. The per-component absolute lag shift  $\delta\Delta\tau_i = |\bar{\Delta\tau}_i^{\text{pert}} - \bar{\Delta\tau}_i^{\text{ctrl}}|$  is compared between components associated with known target genes (identified via highest absolute PCA loading) and all other components using a one-sided Mann-Whitney  $U$  test.